

ADVANCED MACHINE LEARNING · WORKSHEET 11

Dimensionality Reduction and PCA

Feature Extraction · Covariance · Eigenvectors · Principal Components

Lecture 11 — **TEACHER COPY**

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PART I: WHY DIMENSIONALITY REDUCTION?

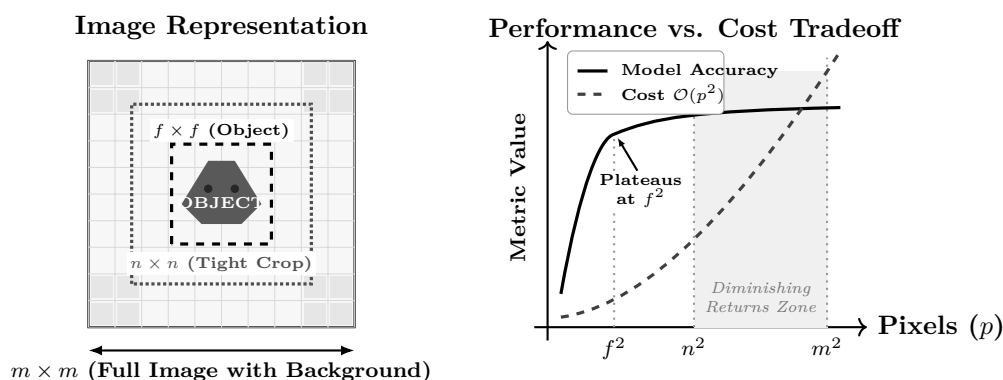
When a dataset has many features, not all of them are equally useful. Some carry important signal while others add noise and computational cost. How do we keep what matters?

Many pixels, little additional signal▷ **HOOK**

In Lecture 10, we saw that adding more features can increase variance and harm generalization. Feature selection answered: “Which original columns should remain?” But what if the information is spread across several columns and no single column captures it alone?

Suppose we are building an AI model that identifies the shape of an object present in an image. The image has size $m \times m$ pixels. Consider the following question:

- Should we feed the complete $m \times m$ image to the model?
- Should we instead use a tightly cropped $n \times n$ image?
- Or should we use a perfectly cropped $f \times f$ image that contains only the object?



Left: An $m \times m$ image with a central object and crop regions. Right: Model performance saturates once the object is captured (f^2), while computational cost continues to explode as uninformative background pixels (m^2) are included.

Classroom question: If the full $m \times m$ image gives nearly the same accuracy as the $f \times f$ crop but requires far more computation, what are the additional pixels contributing?

[Sample Answer]

The additional background pixels mostly contribute random noise, redundant context, and uninformative variation. They exponentially increase computational cost $\mathcal{O}(p^2)$ without providing discriminative signal about the object shape.

SECTION 1 From selecting features to extracting directions

1.1 Why dimensionality can become a learning problem

When a dataset has many features (columns), each observation can be thought of as a point in a high-dimensional space. More features mean more information for the model to process. However, not all features are equally useful. Some features may carry important patterns (signal), while others may add unnecessary complexity or random noise. The challenge is: how do we keep the useful information while reducing the unnecessary parts?

◆ TAKEAWAY

High dimensionality is not automatically bad; the difficulty arises when representation size grows faster than stable information.

1.2 Feature selection and feature extraction

Visual: Selection retains original axes; PCA constructs new axes

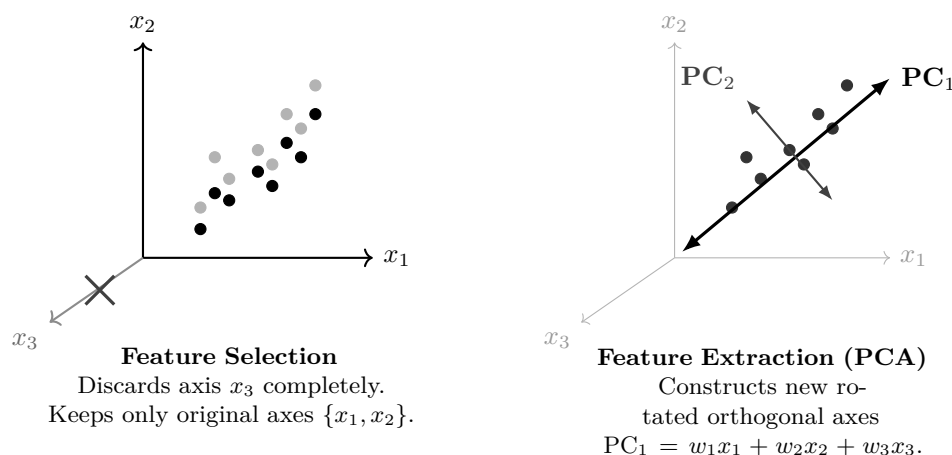


Fig. 1: Feature selection discards entire original features (axes), potentially losing signal. PCA constructs new rotated orthogonal axes that align with the true directions of maximum variance.

1.3 What is PCA?

What is PCA?

Imagine you have a dataset with many features, but most of the useful information is concentrated along just a few directions. PCA is a technique that finds these important directions in the data.

What problem does it solve?

When we have too many features, our models can become slow and may learn noise instead of real patterns. PCA reduces the number of features while keeping as much useful information as possible.

What is it actually doing to the data?

PCA rotates the coordinate axes of the data so that the new axes point along the directions where the data varies the most. It then keeps only the top few directions (called *principal components*) and discards the rest. Think of it as finding the best angle to look at the data so that you see the most spread.

★ KEY INSIGHT

Principal Component Analysis (PCA) is an unsupervised linear feature-extraction method that finds a new set of orthogonal axes (principal components) along which the variance of the data is maximized.

PART II: BUILDING BLOCKS FOR PCA

Before applying PCA, we need to understand a few key concepts: centering, projections, covariance, and eigenvectors.

SECTION 2 Building blocks for PCA

2.1 Data matrix and notation

Each row of X is one observation. Each column is one measured feature. PCA changes the coordinate system used to describe the same observations.

Symbol	Dimensions	Meaning
n	—	Number of observations
p	—	Number of original features
k	—	Number of retained principal components ($k \leq p$)
X	$n \times p$	Original data matrix; row i is observation $\mathbf{x}_i^\top$
$\boldsymbol{\mu}$	$p \times 1$	Vector of training-feature means
X_c	$n \times p$	Centered data matrix
S	$p \times p$	Sample covariance matrix of the centered features
$\mathbf{v}_j$	$p \times 1$	Unit eigenvector / direction of principal component j
λ_j	—	Eigenvalue associated with $\mathbf{v}_j$; variance captured
W_k	$p \times k$	Matrix whose columns are the first k eigenvectors
Z	$n \times k$	Transformed data: the principal-component scores

△ WARNING

Split first. Fit the means, standard deviations, and PCA directions on the **training set only**. Apply the same fitted transformation to validation and test data.

2.2 Step 1: Mean centering

The first step before applying PCA is to **mean center** the data. This means subtracting the average value of each feature from every observation:

$$X_c = X - \mathbf{1}\boldsymbol{\mu}^\top$$

After centering, every feature column has a mean of zero.

If features have **different units** (e.g. rupees vs. years), we should also **standardize** each feature by dividing by its standard deviation:

$$z_{ij} = \frac{x_{ij} - \mu_j}{s_j}$$

- Use **centering only** when features share comparable units and their original variances carry meaningful information.
- Use **standardization** when features have different units or no scale should dominate.

2.3 When one direction captures most of the variance

Axis-aligned variance: Dropping x_2 preserves $> 98\%$ of total variance. Simple **Feature Selection** is sufficient!

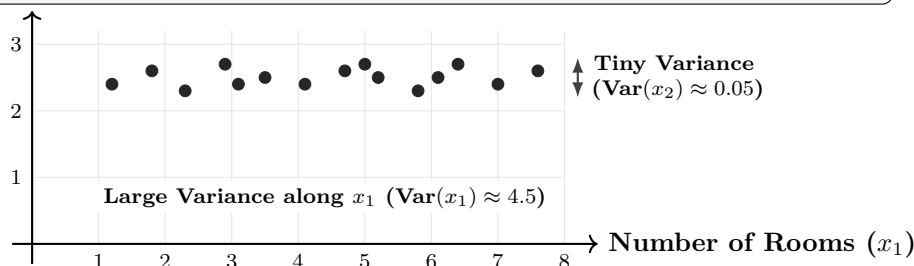


Fig. 2: When the data's variance aligns with an original feature axis, selecting that feature preserves nearly all information.

In this simple example, the data varies much more along the *Number of Rooms* direction than along the *Number of Grocery Shops* direction. If we had to choose only one feature, picking Number of Rooms would preserve most of the information.

But what happens when the data does not align neatly with any original axis?

2.4 Rotating the axes when data lies diagonally

Consider two housing measurements: bedroom area and washroom area. When both measurements increase together, the data cloud lies diagonally. Neither original axis follows the direction in which the data vary most.

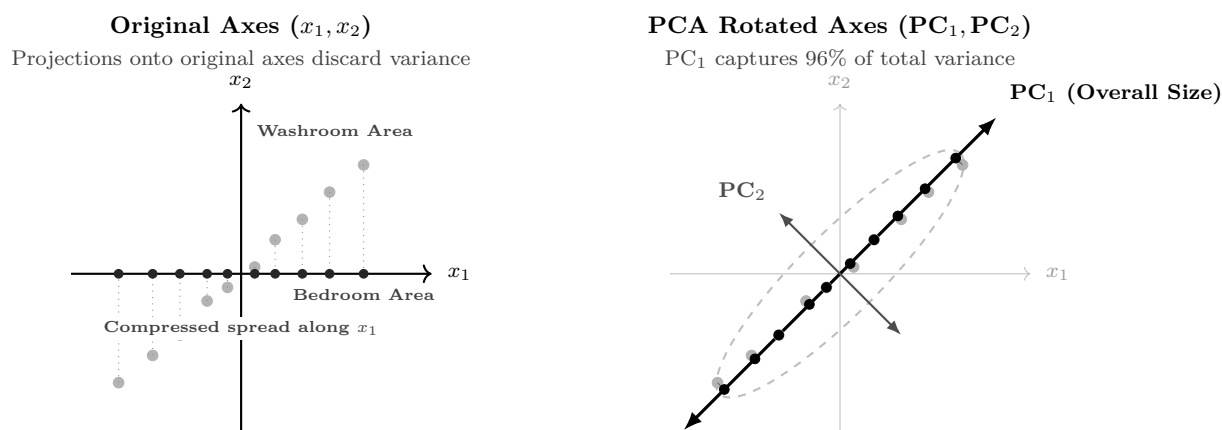


Fig. 3: Left: When features are correlated, projecting onto either original axis (x_1 or x_2) discards valuable variance. Right: PCA rotates the axes to align PC_1 directly along the diagonal spread, capturing nearly all information with a single component.

Unlike the previous example, neither original axis captures the main direction of spread. PCA finds the direction of maximum variance (PC_1) and rotates the coordinate system accordingly.

After this rotation, the entire dataset is transformed into the new coordinate system. If we keep only PC_1 , we reduce two features to one while retaining most of the information.

Important: The number of principal components is always less than or equal to the number of original features.

Geometric interpretation:

- PC_1 can represent an “overall house size” direction because bedroom and washroom area rise together.
- PC_2 is perpendicular and can represent the smaller contrast between how space is allocated to the two rooms.

2.5 Problem formulation: projection and variance

Let $\mathbf{u}$ be any direction in the original p -dimensional space. To use $\mathbf{u}$ as an axis, first make it a **unit vector** so that $\|\mathbf{u}\| = 1$. The scalar coordinate of centered observation $\mathbf{x}_i$ along $\mathbf{u}$ is:

$$z_i = \mathbf{u}^\top \mathbf{x}_i$$

Student exercise: Prove that the scalar coordinate of $\mathbf{x}_i$ along the unit direction $\mathbf{u}$ is given by the dot product $\mathbf{u}^\top \mathbf{x}_i$. Use the definition of projection onto a unit vector.

[Sample Answer]

By definition of vector projection, the orthogonal projection of vector $\mathbf{x}_i$ onto the line spanned by vector $\mathbf{u}$ is given by:

$$\text{proj}_{\mathbf{u}}(\mathbf{x}_i) = \frac{\mathbf{u}^\top \mathbf{x}_i}{\|\mathbf{u}\|^2} \mathbf{u}$$

Since $\mathbf{u}$ is a unit vector, $\|\mathbf{u}\| = 1$ and $\|\mathbf{u}\|^2 = 1$. Thus,

$$\text{proj}_{\mathbf{u}}(\mathbf{x}_i) = (\mathbf{u}^\top \mathbf{x}_i) \mathbf{u} = z_i \mathbf{u}$$

The scalar coordinate along direction $\mathbf{u}$ is therefore $z_i = \mathbf{u}^\top \mathbf{x}_i$.

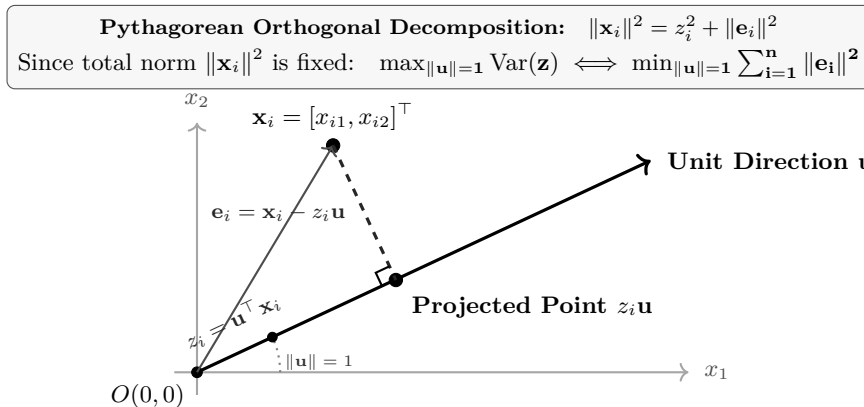


Fig. 4: The dot product $z_i = \mathbf{u}^\top \mathbf{x}_i$ assigns a 1D scalar coordinate to observation $\mathbf{x}_i$ along unit direction $\mathbf{u}$. Maximizing the variance of z_i is mathematically equivalent to minimizing the reconstruction error $\|\mathbf{e}_i\|^2$.

Suppose our centered dataset contains n observations: $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$.

When we project each observation onto the direction $\mathbf{u}$, we get n projection lengths:

$$\mathbf{u}^\top \mathbf{X}_1, \mathbf{u}^\top \mathbf{X}_2, \dots, \mathbf{u}^\top \mathbf{X}_n$$

The variance of these projection lengths measures how spread out the data is along $\mathbf{u}$:

$$\text{Variance along } \mathbf{u} = \frac{1}{n} \sum_{i=1}^n (\mathbf{u}^\top \mathbf{X}_i - \overline{\mathbf{u}^\top \mathbf{X}})^2$$

where $\overline{\mathbf{u}^\top \mathbf{X}}$ is the mean of all projected values.

Among the infinitely many possible unit vectors $\mathbf{u}$, our objective is to find the one that **maximizes** this variance. This is the core problem that PCA solves.

Solving this optimization problem mathematically is beyond the scope of this course. However, the solution turns out to be remarkably elegant: the optimal directions are the **eigenvectors** of the **covariance matrix**, and the corresponding **eigenvalues** tell us how much variance each direction captures.

Before we can apply this result, let us briefly review these key concepts.

2.6 Refresher: covariance matrix, eigenvalues, and eigenvectors

The Covariance Matrix

The **covariance matrix** S is a $p \times p$ matrix that summarizes how every pair of features varies together.

Anatomy of a three-feature covariance matrix:

	f_1	f_2	f_3
f_1	$\text{Var}(f_1)$	$\text{Cov}(f_1, f_2)$	$\text{Cov}(f_1, f_3)$
f_2	$\text{Cov}(f_2, f_1)$	$\text{Var}(f_2)$	$\text{Cov}(f_2, f_3)$
f_3	$\text{Cov}(f_3, f_1)$	$\text{Cov}(f_3, f_2)$	$\text{Var}(f_3)$

- **Diagonal entries** S_{jj} : the variance of each feature.
- **Off-diagonal entries** S_{jk} : the covariance between features j and k .
- **Symmetry**: $S_{jk} = S_{kj}$, so the matrix mirrors across the diagonal.

Eigenvalues and Eigenvectors

For a square matrix A , an **eigenvector** $\mathbf{v}$ is a non-zero vector whose direction is preserved under the transformation — it only gets scaled by a factor λ (the **eigenvalue**):

$$A\mathbf{v} = \lambda\mathbf{v}$$

The eigenvector $\mathbf{v}$ points in a direction that the matrix A does not rotate. The eigenvalue λ tells us how much the matrix stretches or shrinks that direction.

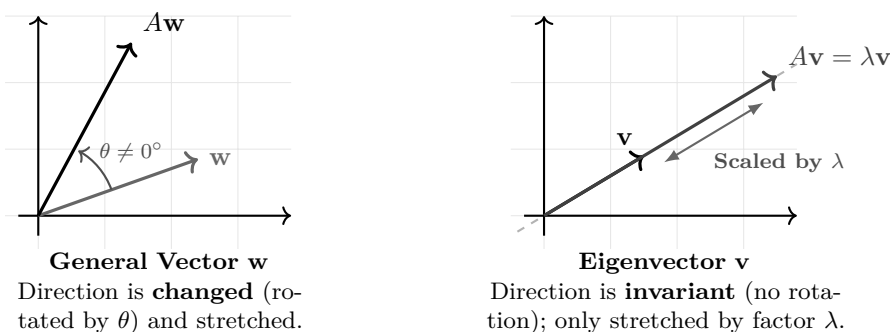


Fig. 5: Unlike a general vector which rotates under a matrix transformation A , an eigenvector $\mathbf{v}$ stays along its original line and is merely scaled by its eigenvalue λ .

Connection to PCA: In PCA, the matrix A is the covariance matrix S . The eigenvectors of S give us the principal directions, and the eigenvalues tell us the variance captured along each direction.

PART III: UNDERSTANDING COVARIANCE

Variance describes how one feature spreads around its mean. Covariance extends the idea to two features.

SECTION 3 Understanding covariance

3.1 Covariance: how two centered features move together

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

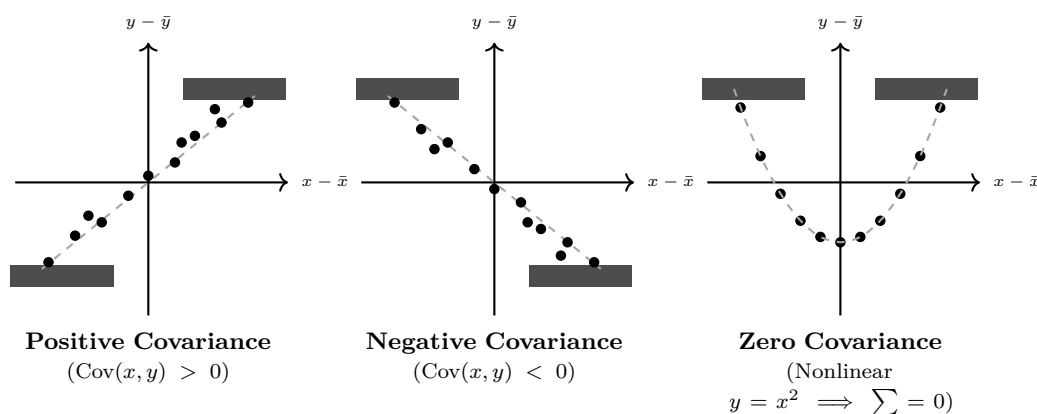


Fig. 6: The sign of covariance indicates linear co-movement. Right: A symmetric parabola has zero linear covariance ($\text{Cov}(x, y) = 0$) despite a perfect deterministic nonlinear relationship, showing that zero covariance does not imply independence.

- **Positive:** Above-average x values tend to appear with above-average y values.
- **Negative:** Above-average x values tend to appear with below-average y values.
- **Near-zero:** Little linear co-movement. A nonlinear relationship can still exist.

Covariance depends on units and is not bounded between -1 and $+1$. Correlation divides covariance by the two standard deviations.

Classroom question: Does zero covariance prove that two variables are independent?

[Sample Answer]

No. Covariance only measures **linear** co-movement. Two variables can have a perfect deterministic nonlinear relationship (e.g., $y = x^2$ symmetric around zero) and have $\text{Cov}(x, y) = 0$.

PART IV: FINDING THE PRINCIPAL COMPONENTS

Now that we understand the building blocks, we can see how PCA finds the principal components.

SECTION 4 Finding the principal components

4.1 Eigenvalues and eigenvectors of the covariance matrix

The principal components are obtained by finding the eigenvalues and eigenvectors of the covariance matrix S . These satisfy:

$$S\mathbf{v} = \lambda\mathbf{v}$$

or equivalently, the eigenvalues can be found by solving:

$$|S - \lambda I| = 0$$

where S is the covariance matrix, λ is an eigenvalue, $\mathbf{v}$ is the corresponding eigenvector, and I is the identity matrix.

4.2 Ordering the principal components

Suppose for a dataset with 3 features, we compute the covariance matrix and obtain:

- Eigenvalues: $\lambda_1, \lambda_2, \lambda_3$
- Corresponding eigenvectors: $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

We sort the eigenvalues in **descending order**. For example, if $\lambda_1 > \lambda_3 > \lambda_2$, then:

Principal Component	Direction	Variance Captured
PC ₁	$\mathbf{v}_1$	λ_1 (largest)
PC ₂	$\mathbf{v}_3$	λ_3 (second largest)
PC ₃	$\mathbf{v}_2$	λ_2 (smallest)

The eigenvector with the largest eigenvalue becomes PC_1 because it captures the most variance.

4.3 Selecting and projecting

The next step is to select the top k principal components that capture most of the data's variance.

To transform each observation into the new coordinate system, we project it onto the selected principal components:

$$z_i = \mathbf{u}^\top \mathbf{X}_i$$

where $\mathbf{u}$ is the unit eigenvector (principal direction) and $\mathbf{X}_i$ is the centered observation.

This projection is performed for **every observation** in the dataset and for **each retained principal component**.

★ KEY INSIGHT

Each eigenvalue of the covariance matrix tells us exactly how much variance is captured along its corresponding principal direction. Larger eigenvalue = more information retained.

4.4 A small numerical example

Three houses described by two already comparable measurements:

House	Bedroom-area index f_1	Washroom-area index f_2
A	2	1
B	4	3
C	6	5

Step 1 — Center the columns. The mean vector is $\boldsymbol{\mu} = [4, 3]^\top$:

$$X_c = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

Step 2 — Compute the sample covariance matrix:

$$S = \frac{1}{2} X_c^\top X_c = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

Step 3 — Find eigenvalues and eigenvectors. Solve $|S - \lambda I| = 0$:

Component	Unit eigenvector	Eigenvalue	Meaning
PC_1	$\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\lambda_1 = \underline{\textcolor{red}{8}}$	Both features increase together.
PC_2	$\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$	$\lambda_2 = \underline{\textcolor{red}{0}}$	No remaining contrast variation.

Step 4 — Project onto PC_1 :

$$\mathbf{z}_1 = X_c \mathbf{v}_1 = \begin{bmatrix} -2\sqrt{2} \\ 0 \\ 2\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} -2.83 \\ 0 \\ 2.83 \end{bmatrix}$$

Step 5 — Interpret. PC_1 captures $\frac{8}{8+0} = \underline{\textcolor{red}{100\%}}$ of the total variance. The two features can be represented by one component with no reconstruction loss because all points lie on a single straight line ($y = x - 1$)

Classroom question: What real-world phrase could describe PC_1 in this example?

[Sample Answer]

“Overall house size” or “Total living scale” (since bedroom and washroom area scale in lockstep).

PRACTICE P1 · FOUNDATIONS AND GEOMETRY

- (a) **Selection or Extraction?** For each scenario, decide whether the problem calls for **feature selection** or **PCA / feature extraction**.

Scenario	Decision	Reason
CGPA_copy exactly duplicates CGPA.	<u>Selection</u>	<u>Exact duplicate column</u>
Bedroom area and washroom area are strongly correlated and both reflect overall house size.	<u>PCA / Extraction</u>	<u>Combine correlated signal</u>
A rare binary medical flag has low variance but is strongly associated with the target.	<u>Selection</u>	<u>PCA would discard it</u>
A 500-dim image embedding must be plotted for human inspection.	<u>PCA / Extraction</u>	<u>Compress to 2D/3D</u>

- (b) **Read covariance from geometry.** For each description, state the expected sign of covariance.

Description	Expected sign	Explanation
Study hours $\uparrow \Rightarrow$ practice-test score $\uparrow$	<u>Positive (+)</u>	<u>Both variables rise together</u>
Product price $\uparrow \Rightarrow$ units sold $\downarrow$	<u>Negative (-)</u>	<u>One rises as other falls</u>
Points form a symmetric U-shape around $x = 0$.	<u>Near Zero (0)</u>	<u>Symmetric cancellation</u>

PRACTICE P2 · MATHEMATICAL CONCEPTS

- (a) **Compute one projection.** Let $\mathbf{x} = [3, 1]^\top$ and $\mathbf{u} = \frac{1}{\sqrt{2}}[1, 1]^\top$.

(i) Verify that $\mathbf{u}$ is a unit vector: $\sqrt{(\frac{1}{\sqrt{2}})^2 + (\frac{1}{\sqrt{2}})^2} = \sqrt{1/2 + 1/2} = 1$

(ii) Compute the score: $z = \mathbf{u}^\top \mathbf{x} = \frac{1}{\sqrt{2}}(3) + \frac{1}{\sqrt{2}}(1) = \frac{4}{\sqrt{2}} = 2\sqrt{2} \approx 2.828$

(iii) Compute the projected point: $z\mathbf{u} = 2\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

(iv) Explain why the projected point does not have to equal $\mathbf{x}$:

[Sample Answer]

Because $\mathbf{x} = [3, 1]^\top$ does not lie exactly along the line spanned by $\mathbf{u} = \frac{1}{\sqrt{2}}[1, 1]^\top$. The residual difference $\mathbf{e} = \mathbf{x} - z\mathbf{u} = [1, -1]^\top$ is orthogonal to $\mathbf{u}$ and represents the information lost during 1D projection.

- (b) **Complete a miniature PCA.** The centered observations are $\mathbf{x}_1 = [-1, -1]^\top$, $\mathbf{x}_2 = [0, 0]^\top$, $\mathbf{x}_3 = [1, 1]^\top$.

- (i) Form X_c and compute $S = \frac{1}{n-1} X_c^\top X_c$:

$$X_c = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}, \quad S = \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

- (ii) Show that $\mathbf{v}_1 = \frac{1}{\sqrt{2}}[1, 1]^\top$ has eigenvalue $\lambda_1 = 2$ ($S\mathbf{v}_1 = 2\mathbf{v}_1$)
 (iii) Show that $\mathbf{v}_2 = \frac{1}{\sqrt{2}}[1, -1]^\top$ has eigenvalue $\lambda_2 = 0$ ($S\mathbf{v}_2 = 0\mathbf{v}_2$)
 (iv) Explained variance ratio of PC₁: $\frac{2}{2+0} = 100\%$. Loss? **Yes, completely lossless.**

PART V: THE COMPLETE PCA PROCEDURE AND PRACTICAL DECISIONS

PCA has been explained. Now organize the procedure end-to-end and address choosing k .

SECTION 5 The complete PCA procedure and practical decisions

5.1 Step-by-step PCA workflow

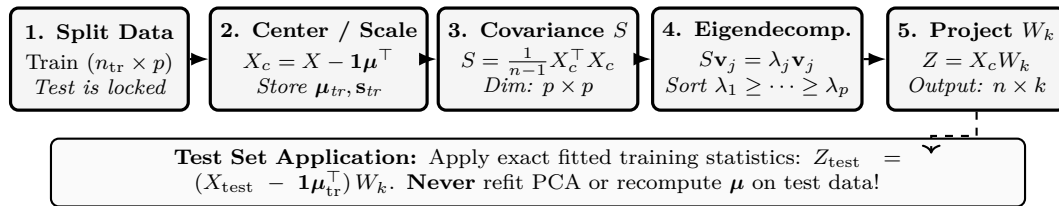


Fig. 7: The complete end-to-end PCA machine learning pipeline. All statistics (μ, s) and projection directions (W_k) are learned exclusively on the training set and applied unchanged to test/production data.

1. Create the data split before learning any preprocessing. The test set remains untouched.
2. Fit centering (and, when appropriate, standardization) on the **training features**. Store μ and s .
3. Construct the covariance matrix S from the centered training data.
4. Find eigenvalues and unit eigenvectors of S .
5. Sort eigenpairs by descending eigenvalue.
6. Choose k using explained variance, visualization needs, and downstream validation.
7. Form $W_k = [\mathbf{v}_1 \cdots \mathbf{v}_k]$. Its dimensions are $p \times k$.
8. Transform: $Z_{\text{train}} = X_{c,\text{train}} W_k$ and $Z_{\text{test}} = X_{c,\text{test}} W_k$, using **stored training** quantities.

5.2 How do we determine the optimum number of principal components?

Recall that each eigenvalue λ_j of the covariance matrix represents the **amount of variance** captured by the corresponding principal component. A large eigenvalue means that direction captures a lot of the data's spread; a small eigenvalue means that direction captures very little.

The **total variance** in the dataset is simply the sum of all eigenvalues:

$$\text{total variance} = \sum_{j=1}^p \lambda_j = \text{trace}(S)$$

The **Explained Variance Ratio (EVR)** tells us what fraction of the total variance each principal component captures:

$$\text{EVR of PC}_j = \frac{\lambda_j}{\sum_{\ell=1}^p \lambda_{\ell}}, \quad \text{cumulative EVR through } k = \frac{\sum_{j=1}^k \lambda_j}{\sum_{\ell=1}^p \lambda_{\ell}}$$

Visual: Scree plot with cumulative explained variance

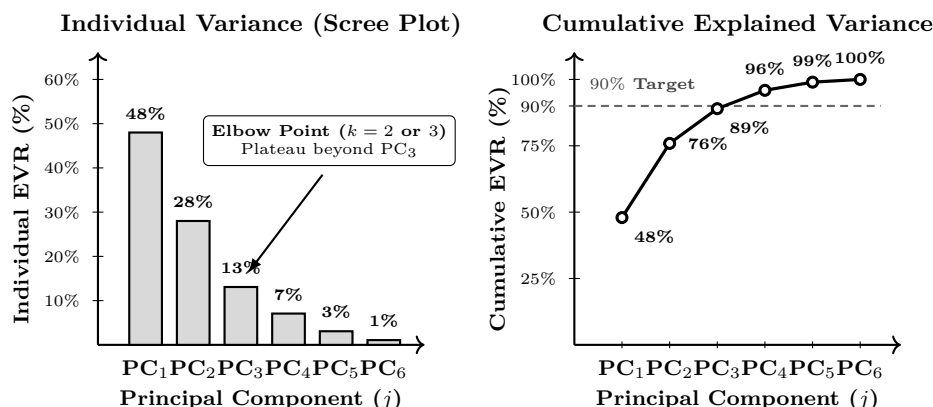


Fig. 8: A scree plot displays individual eigenvalues (bars) and cumulative explained variance (line). The “elbow” indicates where additional components yield diminishing returns.

Rules for choosing k :

- **Cumulative threshold:** Choose the smallest k reaching 90% or 95%, then validate.
- **Elbow / scree rule:** Look for the point after which eigenvalues flatten.
- **Visualization:** Choose $k = 2$ or 3 for human inspection.
- **Downstream prediction:** Treat k as a hyperparameter inside cross-validation.

Classroom question: If the first two components capture 76% of total variance, is $k = 2$ automatically the correct choice?

[Sample Answer]

Not automatically. If the downstream task requires fine distinctions that reside in PC₃ (even if PC₃ only accounts for 13% of variance), choosing $k = 2$ will cause underfitting. k should be tuned using cross-validation on the actual prediction task.

5.3 When PCA does not perform well

PCA works by finding directions of maximum variance. However, there are situations where this approach fails:

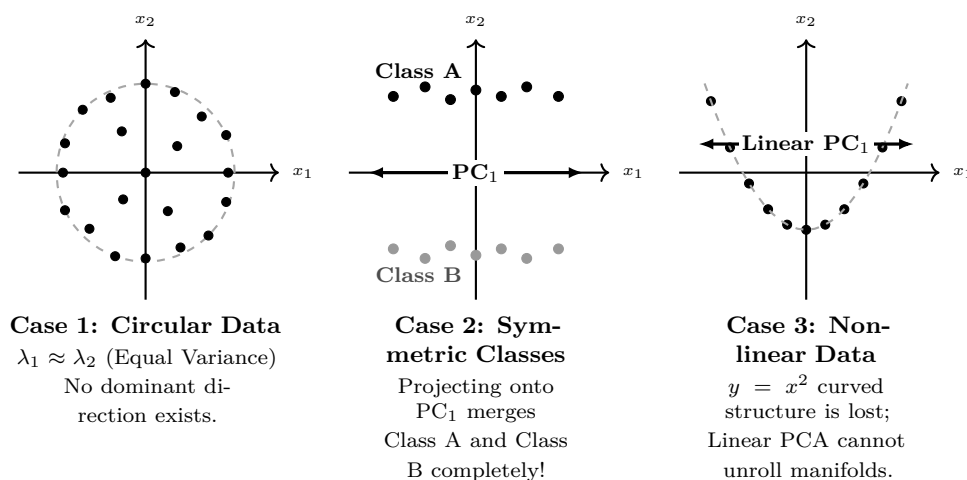


Fig. 9: Key failure cases of linear PCA. Left: Isotropic spread yields arbitrary components. Middle: Unsupervised variance maximization can destroy class separability. Right: Linear projections cannot capture curved nonlinear manifolds.

5.4 What PCA changes — and what it can fail to preserve

PCA can help with:

- Visualization and compression
- Faster downstream learning
- Removal of linear redundancy
- Noise reduction when low-variance directions are mostly noise

PCA does not guarantee:

- Better prediction
- Causal meaning or class separation
- Preservation of a low-variance feature that is highly predictive of y

△ WARNING

Do not fit scaling or PCA before cross-validation and then cross-validate the downstream model. Fit PCA inside each fold on the training portion only.

5.5 Comparison with feature selection

Criterion	Feature Selection	Feature Extraction (PCA)
What is reduced?	Number of original columns retained	Number of new component coordinates
Are original meanings preserved?	Yes	Usually not directly
Treatment of correlation	Often keeps one representative, removes another	Combines shared variation into one direction
Uses the target y ?	Depends on the method	No — standard PCA is unsupervised
Primary mathematical idea	Keep or discard columns	Rotate the coordinate system, then project

◆ TAKEAWAY

Feature selection asks which original axes deserve to remain. PCA asks whether the apparent complexity of the feature space can be summarized by a smaller set of rotated axes that still capture the relevant structure.

PRACTICE P3 · DIAGNOSIS AND APPLICATION

(a) **Choose k from eigenvalues.** Suppose the eigenvalues are $[6, 2, 1, 0.5]$.

- (i) Compute total variance $6 + 2 + 1 + 0.5 = 9.5$
 (ii) Complete the table:

k	$\sum_{j=1}^k \lambda_j$	Cumulative ratio	$\geq 90\%$?
1	<u>6.0</u>	$6.0/9.5 = 63.2\%$	<u>No</u>
2	<u>8.0</u>	$8.0/9.5 = 84.2\%$	<u>No</u>
3	<u>9.0</u>	$9.0/9.5 = 94.7\%$	<u>Yes</u>
4	<u>9.5</u>	$9.5/9.5 = 100\%$	<u>Yes</u>

- (iii) Smallest k that retains at least 90% **(captures 94.7%)**
 (iv) Explain why that k is still only a candidate for a predictive pipeline:

[Sample Answer]

Variance does not equate to predictive relevance. The dropped direction (PC_4 , carrying only 5.3% of variance) might happen to contain the critical signal distinguishing the target y . Thus, k must be cross-validated against prediction error.

(b) **Diagnose the pipeline.** A student standardizes the full dataset, fits PCA once using all observations, keeps the first five components, and then reports five-fold cross-validation accuracy.

- (i) Is the reported score trustworthy? **No, it is overly optimistic.**
 (ii) Which steps have seen the validation folds?

[Sample Answer]

Both standardization (means μ and standard deviations s) and PCA (the covariance matrix S and eigenvectors W_k) were computed using the entire dataset including validation samples, resulting in data leakage.

- (iii) Describe the correct fold-by-fold procedure:

[Sample Answer]

In each CV fold: (1) Compute means μ_{train} and scales s_{train} on the training split only; (2) Center and scale train data; (3) Compute S_{train} and eigenvectors W_k on train data; (4) Transform train data $Z_{\text{train}} = X_{c,\text{train}} W_k$; (5) Transform validation fold using stored train parameters: $Z_{\text{val}} = (X_{\text{val}} - \mu_{\text{train}}) W_k$; (6) Train model on Z_{train} and evaluate on Z_{val} .

PRACTICE P4 · COMPREHENSIVE CHALLENGE

(Homework)

(a) **Fill in the Key Terms table:**

Term	Definition
Centering	Subtracting feature means so that each feature column has mean zero.
Covariance matrix	Symmetric matrix containing pairwise covariances and feature variances.
Eigenvector	whose direction is invariant under matrix multiplication: $A\mathbf{v} = \lambda\mathbf{v}$.
Scalar factor	by which an eigenvector is scaled; equals variance along that principal component.
Principal component	An orthogonal coordinate axis pointing along a direction of successive maximum variance.
Explained variance	Fraction of total dataset variance accounted for by a given principal component: $\lambda_j / \sum \lambda_\ell$.
Scree plot	A line graph of sorted eigenvalues used to visually detect the “elbow” for selecting k .

(b) **True or False:**

Statement	T/F	Justification
PCA uses the target variable y to find directions.	[F]	PCA is strictly unsupervised.
Principal components are always unrelated.	[T]	Eigenvectors of symmetric S are orthogonal.
Standardization is always necessary before PCA.	[F]	Only when units or scales differ arbitrarily.
The eigenvalue of a PC equals the variance of the scores along that direction.	[T]	$\text{Var}(\mathbf{z}_j) = \mathbf{v}_j^\top S \mathbf{v}_j = \lambda_j$.
PCA can unfold a curved manifold.	[F]	Standard PCA is strictly linear.

(c) **Synthesis questions.** Answer any three:

[Instructor Guidance / Solution]

(i) **Centering:** Ensures that coordinates represent spread around the mean rather than distance from an arbitrary origin, making $X_c^\top X_c / (n - 1)$ the true covariance matrix.

(ii) **Geometric eigenvalue:** λ_j represents the variance (squared spread) of the data points when projected onto unit vector $\mathbf{v}_j$. The square root $\sqrt{\lambda_j}$ gives the semi-axis length of the data scatter ellipse.

(iii) **Uncorrelated scores:** The covariance matrix of the transformed data $Z = X_c W$ is $W^\top S W = \Lambda$ (a diagonal matrix of eigenvalues). Off-diagonal covariances are all zero.

(iv) **Standardization impact:** When features have vastly different numeric scales (e.g., income in dollars vs. age in years), the unstandardized covariance matrix is dominated entirely by the large-scale variable. Standardizing gives all features equal initial weight.

(v) **Improved vs. harmed models:** PCA helps when multicollinearity causes high variance in linear models, or when low-variance components are pure noise. It harms performance when the target y happens to correlate with a low-variance component that PCA discards, or when class boundaries are nonlinear.

(vi) **Bias–variance tradeoff:** Retaining fewer principal components ($k < p$) reduces model variance and multicollinearity at the cost of introducing approximation bias (discarding $(p - k)$ directions of variation).

(vii) **Failure example:** A dataset where points lie on a 1D spiral or concentric circles in 2D space. Linear PCA projects the spiral onto a line, crushing distinct arms on top of each other.

BY THE END OF THIS SHEET, YOU CAN

- [✓] Distinguish feature selection from feature extraction and state when PCA is appropriate.
- [✓] Write the centering formula and explain when standardization is needed.
- [✓] Explain what the covariance matrix records and identify its diagonal and off-diagonal entries.
- [✓] Explain how eigenvalues and eigenvectors of the covariance matrix yield the principal components.
- [✓] State why the principal directions are eigenvectors of the covariance matrix and why each eigenvalue represents the variance captured.
- [✓] Project data onto a unit direction and interpret the score geometrically.
- [✓] Complete a miniature PCA calculation by hand.
- [✓] Use explained variance ratios and a scree plot to choose k .
- [✓] Identify situations where PCA does not perform well.
- [✓] Explain why PCA must occur inside cross-validation folds.
- [✓] Compare PCA with feature selection across purpose, supervision, and validation discipline.